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Q1 **Harshad number**

10 marks

A number is called a Harshad number (or Niven number) if it is divisible by the sum of its digits. Example: 18 → digit sum = 9, 18 % 9 == 0. Write a Python program to read a number N from the user and check whether it is a Harshad number. Also print all Harshad numbers between 1 and 500.

Your Answer

```
# Function to check Harshad number
def is_harshad(num):
    temp = num
    digit_sum = 0

    while temp > 0:
```

Save Answer

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Q2 **GCD**

10 marks

The following Python program is intended to find the GCD (Greatest Common Divisor) of two numbers using the Euclidean algorithm. It contains FOUR errors. Identify each error and write the corrected line.

a = int(input("Enter first: ")) b = int(input("Enter second: "))

while b = 0: temp = a a = b b = temp % a

print("GCD is", a)

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
a = int(input("Enter first: "))
b = int(input("Enter second: "))

while b != 0:
    temp = b
    b = a % b
    a = temp

print("GCD is", a)
```

Test Input

```
24
18
```

Output

```
Enter first: Enter second: GCD is 6
```

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```
def find_pairs(lst, target):
seen = _____ pairs = []
for num in _____:
complement = target - _____
if complement in _____:
pairs.append((complement, num))
_____.add(num)
return _____
data = [int(x) for x in input("Enter numbers: ").split()]
t = int(input("Target: "))
result = _____(data, t)
print("Pairs:", result)
```

Your Answer

```
def find_pairs(lst, target):
  seen = set()
  pairs = []

  for num in lst:
    complement = target - num
```

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Q4 **Bitwise Operators** 10 marks

Trace the following Python program step by step. Write the exact output. Also state the data type of each variable after assignment.

```
a = 0b1101
b = 0o25
c = 0x1A
result = (a & b) | c
print(f"a={a}, b={b}, c={c}, result={result}")
print(bin(result), oct(result), hex(result))
print(result >> 2, result << 1, ~result)
```

Your Answer

a=13, b=21, c=26, result=31
0b11111 0o37 0x1f
7 62 -32

Data Types:
a → int

Save Answer

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t = input("Target: ")
result = rotated_binary_search(data, t)
print("Found at:", result)

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
def rotated_binary_search(arr, target):
    low = 0
    high = len(arr) - 1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == target:
            return mid

        if arr[low] <= arr[mid]:
            if target >= arr[low] and target < arr[mid]:
                high = mid - 1
```

Test Input

```
6
```

Output

```
Target: Found at: 2
```